Learning from Waiting

1. Background and purpose

We designed an in-person wait-time decomposition experiment to study whether consumers use queuing time to infer product quality. Total waiting time is decomposed into queuing time and shipping time. Shipping is designed to be a pure waiting cost; queuing may additionally convey information through earlier consumers' purchase decisions.

The comparison between Q50 and the Q00 benchmark asks whether consumers respond to the informational content of waiting, beyond its total cost. An earlier purchase is a positive quality signal; an earlier nonpurchase is a negative quality signal in the proposed learning mechanism. All regressions in this note use only uninformed decisions.

2. Hypotheses

H1. Waiting composition and positive purchase signals

Holding rank and total waiting time fixed, moving one unit from shipping to queuing should increase purchase probability in Q50, with a more positive slope than Q00. The Q00 composition slope is predicted to be negligible. At fixed rank, more earlier purchases also imply fewer earlier nonpurchases.

H2. Rank and negative nonpurchase signals

Holding queuing and shipping fixed, a higher rank should lower purchase probability in Q50, with a more negative slope than Q00. The Q00 rank slope is predicted to be negligible. Given the experimental mapping from purchases to queuing, higher rank at fixed queuing corresponds to more earlier nonpurchases.

3. Joint regression specification

We estimate a pooled linear probability model (OLS) with treatment-specific slopes for queuing time, rank, and total waiting time:

$$Y_{it} = \alpha + \tau T_i + b_q q_{it} + \gamma_q T_i q_{it} + b_r r_{it} + \gamma_r T_i r_{it} + b_d d_{it} + \gamma_d T_i d_{it} + \varepsilon_{it}.$$

Notation	Definition
Y_{it}	Purchase decision: 1 = purchase, 0 = no purchase
T_i	Treatment: 1 = Q50, 0 = Q00
q_{it}, s_{it}	Queuing time and shipping time
$d_{it} = q_{it} + s_{it}$	Total waiting time
r_{it}	Rank
i, t	Participant and paid round
b_q, b_r, b_d	Q00 slopes for the corresponding variables
γ_q, γ_r, γ_d	Q50 minus Q00 slope differences
α, τ, ε_{it}	Q00 intercept, treatment intercept difference, and residual

4. Regression results and hypothesis assessment

Table 1. Joint linear model

The sample includes all 1,650 formal-round uninformed decisions: 800 in Q00 and 850 in Q50, from 100 participants. No common-support filter is applied. Coefficients, standard errors and confidence limits are in percentage points.

Hyp.	Target	Symbol	Coef. (pp)	SE (pp)	95% CI (pp)	Two-sided p	Test df
H1	Q00	b_q	-9.20	5.04	[-19.08, +0.68]	0.068063	1642.00
H1	Q50	$b_q + \gamma_q$	+24.00	3.61	[+16.92, +31.07]	3.973e-11	1642.00
H1	Q50 - Q00	γ_q	+33.19	6.20	[+21.04, +45.35]	9.688e-08	1642.00
H2	Q00	b_r	-3.16	3.05	[-9.14, +2.82]	0.300465	1642.00
H2	Q50	$b_r + \gamma_r$	-17.97	1.69	[-21.28, -14.66]	1.085e-25	1642.00
H2	Q50 - Q00	γ_r	-14.81	3.49	[-21.65, -7.98]	0.000023	1642.00

All reported p-values are two-sided; confidence intervals are two-sided 95% intervals. Q50 - Q00 denotes a slope difference, not an average purchase-rate difference.

H1: waiting composition

The Q50 composition coefficient is +24.00 pp (two-sided p = 3.973e-11). The Q50-Q00 difference is +33.19 pp (two-sided p = 9.688e-08). These results support the predicted positive Q50 composition association and positive treatment difference.

The Q00 estimate is -9.20 pp (two-sided p = 0.068063). It is not statistically distinguishable from zero at the 5% level, but this does not establish a negligible effect.

H2: rank

The Q50 rank coefficient is -17.97 pp (two-sided p = 1.085e-25). The Q50-Q00 difference is -14.81 pp (two-sided p = 0.000023). These results support the predicted negative Q50 rank association and negative treatment difference.

The Q00 estimate is -3.16 pp (two-sided p = 0.300465). It is not statistically distinguishable from zero at the 5% level, but equivalence to zero has not been established.

Interpretation

The findings are consistent with the proposed learning mechanism. Establishing correct quality inference would require an explicit benchmark for beliefs or optimal choices. Negligible Q00 effects require a substantive equivalence margin and a separate test. The estimates are conditional associations, and the reported p-values are exploratory and unadjusted for multiple testing.